# Ender Batu Demirtaş

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## Education

**California State University, Fullerton** — Fullerton, CA

*Bachelor of Science in Computer Science* — *May 2026*

- **Relevant Coursework:** Data Structures, Software Engineering, Operating Systems, OOP, Back-end Development, Machine Learning
- **Certification:** AWS Certified Generative AI Developer – Professional (AIP-C01), exam scheduled November 2026

## Technical Skills

**Programming Languages:** Python, C++, Swift, SQL
**Generative AI:** Amazon Bedrock (Agents, Knowledge Bases, Guardrails), Bedrock AgentCore, Strands Agents, LangGraph, RAG, Vector Search, Prompt Engineering, LLM Evaluation
**Cloud:** AWS (Bedrock, Lambda, API Gateway, Step Functions, S3, RDS, IAM, KMS, Secrets Manager, CloudWatch, App Runner, EC2)
**Frameworks & Libraries:** FastAPI, Pydantic, Streamlit, pandas, scikit-learn, SwiftUI, SwiftData
**Tools & Concepts:** Git, Jira, Confluence, Algorithms, Machine Learning, REST APIs, CI/CD, Data Pipelines, Unit Testing

## Projects

**NL2SQL Chatbot** | *Python, FastAPI, Strands Agents, Amazon Bedrock, AWS* September 2025 – November 2025

- Built a chatbot that lets non-technical users query a **PostgreSQL** database in plain English, using a **Strands** agent on **Amazon Bedrock** (Claude 3.5 Haiku) that calls custom tools to generate and run SQL.
- Served the agent through a **FastAPI** backend and a lightweight web UI, keeping database credentials in **AWS Secrets Manager** instead of code.
- Containerized and deployed the service to **AWS App Runner**, backed by a **PostgreSQL** database on **AWS RDS**.

**Memoria** | *SwiftUI, SwiftData, Swift Charts, macOS + iOS* | Solo Developer January 2026 – July 2026

- Shipped **4 tagged releases** as sole engineer of an offline-first **SwiftData** app for macOS and iOS.
- Engineered a custom accounting engine for per-trade VWAP cost basis, time-weighted return, and daily **P&L** attribution, benchmarked against the **S&P 500** via a REST API.
- Built an interactive equity curve with gesture scrubbing (**Swift Charts**), a P&L calendar heatmap, and a live watchlist with sparklines.

**Catalyst Insight Studio** | *Python, Streamlit, pandas, scikit-learn* | Co-Developer May 2026 – July 2026

- Enabled ∼**300 General Chemistry II students per semester** to explore lab results in the browser with no install, by co-building a **Streamlit** app and owning its **pandas** data pipeline and **scikit-learn** models.
- Turned messy student survey exports into analysis-ready data, by building the normalization, cleaning, and validation pipeline.
- Kept the app working as the lab's data-collection forms evolved, by adding automated schema-mismatch validation and a **New Semester** tab for merging raw Excel exports.
- Eased handoff to the next student team, by breaking up a monolithic app into separate UI and data layers and writing **54 unit tests**.

## Leadership

**Data Science & Machine Learning Club (DSML) – CSUF** — Fullerton, CA

*Vice President* — *January 2025 – May 2026*

- Led weekly presentations and live tool demos on core data science and machine learning concepts, combining theory with real-world applications.
- Designed case study-driven curricula using realistic datasets to mirror production machine learning challenges, delivered in hybrid sessions for online and in-person members.

**Association for Computing Machinery (ACM) – CSUF** — Fullerton, CA

*Artificial Intelligence Board Member* — *January 2025 – May 2026*

- Led an overhaul of presentation and workshop formats, increasing attendance by **30%+**.
- Revamped our engagement strategies, growing activity participation from negligible to **25+ students** in interactive activities.

**Technical Event Production & Outreach** — Fullerton, CA

*Bridging the Gap Between Students and Industry* — *January 2025 – December 2025*

- Founded and coordinated a guest speaker series, hosting engineering leaders from companies including AWS to deliver panels on production architectures and data engineering.
- Attracted **20+ students per session** by promoting interactive discussions on modern cloud and back-end practices.